A simple example which we would wish to include in our framework is the point distribution which concentrates all measure at a single point. This type of object cannot be directly represented by a smooth function (or even by one in L^2), but it is nonetheless possible to extend our framework in a way that makes sense in such cases.

5.3.1 Generalized Functions

In order to address the need to describe the evolution of a broader class of distributions, we introduce the concept of "generalized function" as a mathematical framework incorporating, for example, Dirac delta functions.[4] Formally, we define a *generalized function* as a real-valued linear operation (a *linear functional*) defined on a suitable space of functions. The functions on which the generalized function acts are called *test functions*. Some choices for the space of test functions $\mathscr{T}^m$ defined on $\mathbb{R}^m$ include:

- the C^∞ functions with compact support in $\mathbb{R}^m$, denoted $C_0^\infty(\mathbb{R}^m)$,
- the space $S(\mathbb{R}^m)$ of C^∞ functions on $\mathbb{R}^m$ whose derivatives decay rapidly at infinity (the so-called *Schwarz space*)[5]

If f is a bounded, smooth function on $\mathbb{R}^m$, we may define a generalized function T_f by reference to its *action* on a test function $\phi \in \mathscr{T}$ via integration:

$$T_f(\phi) = \int_{\mathbb{R}^m} f(z)\phi(z)\mathrm{d}\omega.$$

This is very like the L^2 inner product, but the functions involved do not both need to lie in L^2. Nevertheless, we write

$$T_f(\phi) = \langle f, \phi \rangle,$$

[4]The alternate term for "generalized function" is "distribution" which is suggestive of "probability distribution," but "distribution" is a more general concept.

[5]The formal definition of S is

$$S(\mathbb{R}^m) \stackrel{\text{def}}{=} \{f \in C^\infty(\mathbb{R}^m) | \|f\|_{\boldsymbol{\alpha},\boldsymbol{\beta}} < \infty\},$$

where $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ are arbitrary *multi-indices*, i.e. vectors of non-negative integers, and

$$\|f\|_{\boldsymbol{\alpha},\boldsymbol{\beta}} \stackrel{\text{def}}{=} \sup_{z\in\mathbb{R}^m} \left| z_1^{\alpha_1} z_2^{\alpha_2} \cdots z_m^{\alpha_m} \frac{\partial^{\beta_1+\beta_2+\ldots+\beta_m} f}{\partial z_1^{\beta_1} \partial z_2^{\beta_2} \cdots \partial z_m^{\beta_m}} \right|.$$